Model Predictive Control — Spring 2026

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Solution 05

Invariance

Exercise 1 **Largest inscribed ball**

Recall the LP formulation of the problem:

$$\begin{aligned} \min_{x,r} \quad & -r \\ \text{subj. to} \quad & 0 \le x_1 \le 2 \\ & 0 \le x_2 \le 1 \\ & a_i^T x + r\|a_i\| \le b_i, \quad i = 1, \dots, m \end{aligned}$$

Resulting Matrices for problem 1 and 2 in MATLAB (min $f'x$ subj. to $Ax \le b$):

$$\begin{aligned} f &= \begin{bmatrix} 0 & 0 & -1 \end{bmatrix}^\top \\ A_1 &= \begin{bmatrix} 1 & 0 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 2 & 0 & 1 & 0 \end{bmatrix}^\top \\ A_2 &= \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 2 & 1 & \sqrt{5} \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 & 0 & 2 \end{bmatrix}^\top \end{aligned}$$

The results of the computations are depicted in Figure 1. The solution to the first problem is obviously not unique since all circles with a radius of 0.5 and a center position of $0.5 \le x_1 \le 1.5$ and $x_2 = 0.5$ optimize the value function. Problem 2 on the other hand has a unique solution of $x_1 = 0.382$, $x_2 = 0.382$ and a radius of 0.382.

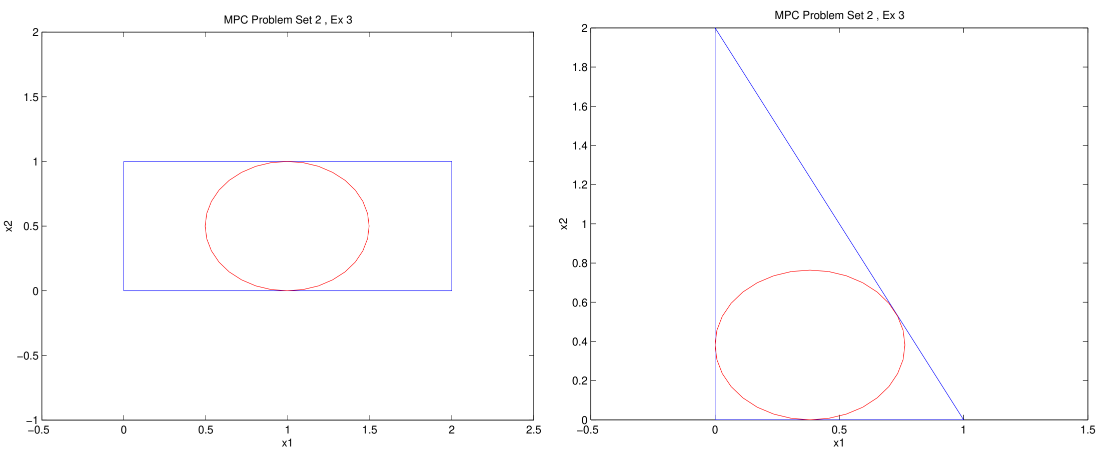

Figure 1: Solutions to the two problems

Exercise 2 **Minkowski sum of invariant sets**

Let $X = aX_1 \oplus (1-a)X_2$ and $x(0)$ in X. The Minkowski sum implies there exist $x_1(0)$ in X_1 and $x_2(0)$ in X_2 such that $x(0) = ax_1(0) + (1-a)x_2(0)$. Using the dynamics $x_1(k+1) = Ax_1(k), x_2(k+1) = Ax_2(k)$, we have by invariance of X_1 and X_2 that $x_1(k)$ in X_1, $x_2(k)$ in X_2 for all k. We also have $x(k) = ax_1(k) + (1-a)x_2(k)$ for all k, which is therefore in the Minkowski sum $X = aX_1 + (1-a)X_2$, proving the claim.

Exercise 3 **Invariance and feedback laws**

F has to fulfill the following two properties:

$$\begin{aligned}&(A+BF)x \in \mathcal{X}_f \quad \forall x \in \mathcal{X}_f\ ,\\ &Fx \in \mathcal{U} \quad \forall x \in \mathcal{X}_f\ ,\end{aligned}$$

as $\mathcal{X}_f$ as well as $\mathcal{U}$ are boxes, and the maps $A+BF$ and F are linear, it is sufficient to check all the corner points of $\mathcal{X}_f$.

Thus we need to check that the two conditions hold for the following points.

$$(10,10) \quad (10,-10) \quad (-10,-10) \quad (-10,10)$$

$$A+BF = \begin{bmatrix} 0.5+f_1 & -0.25 \\ 0.25 & 0.5+f_2 \end{bmatrix}$$

For point $(10,10)$ we have:

$$\begin{aligned}&-10 \leq 10(0.5+f_1-0.25) \leq 10\\ &-10 \leq 10(0.5+0.25+f_2) \leq 10\\ &\Leftrightarrow\\ &-1-(0.5-0.25) \leq f_1 \leq 1-(0.5-0.25)\\ &-1-(0.5+0.25) \leq f_2 \leq 1-(0.5+0.25)\\ &\Leftrightarrow\\ &-1.25 \leq f_1 \leq 0.75\\ &-1.75 \leq f_2 \leq 0.25\end{aligned}$$

Additionally we have the inequalities from the input constraints

$$\begin{aligned}&-0.5 \leq f_1 \leq 0.5\\ &-0.5 \leq f_2 \leq 0.5\end{aligned}$$

Together the resulting inequalities for F at point $(10,10)$ are,

$$\begin{aligned}&-0.5 \leq f_1 \leq 0.5\\ &-0.5 \leq f_2 \leq 0.25\end{aligned}$$

Using the same calculations for point $(10,-10)$ results in,

$$\begin{aligned}&-0.5 \leq f_1 \leq 0.25\\ &-0.5 \leq f_2 \leq 0.5\end{aligned}$$

for $(-10, -10)$

$$
\begin{aligned}
-0.5 \le f_1 \le 0.5 \\
-0.5 \le f_2 \le 0.25
\end{aligned}
$$

for $(-10, 10)$

$$
\begin{aligned}
-0.5 \le f_1 \le 0.25 \\
-0.5 \le f_2 \le 0.5
\end{aligned}
$$

Thus we conclude that, if F is in the following interval,

$$
\begin{aligned}
-0.5 \le f_1 \le 0.25 \\
-0.5 \le f_2 \le 0.25
\end{aligned}
$$

the set $\mathcal{X}_f$ is invariant for the closed loop system under the input constraints.

Exercise 4 **Maximal Invariant Set**

1) Give an algorithm for computing the maximal invariant set.

(a) $\Omega_0 \leftarrow \mathcal{X}$
(b) loop
(c) $\quad \Omega_{i+1} \leftarrow \text{pre}(\Omega_i) \cap \Omega_i$
(d) $\quad$ if $\Omega_{i+1} = \Omega_i$ then
(e) $\quad\quad$ return $\Omega_\infty = \Omega_i$
(f) $\quad$ end if
(g) end loop

2) Compute the maximal invariant set for this system.

$$
\begin{aligned}
\Omega_0 &= \{x \mid -0.5 \le x \le 2\} \\
\text{pre}(\Omega_0) &= \{x \mid -0.5 \le -0.5x \le 2\} = \{x \mid -4 \le x \le 1\} \\
\Omega_1 &= \{x \mid -0.5 \le x \le 1\} \\
\text{pre}(\Omega_1) &= \{x \mid -0.5 \le -0.5x \le 1\} = \{x \mid -2 \le x \le 1\} \\
\Omega_2 &= \{x \mid -0.5 \le x \le 1\} = \Omega_\infty
\end{aligned}
$$